

Introduction to R

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Outline

Introduction

R and R Studio

Basic

Practice

About Me

- ▶ Rex Wang
- ▶ First-year master's student in Agronomy Biostatistics
- ▶ I will not be in Steven's lecture
- ▶ Feel free to email me if you have any questions
- ▶ Email: R14621207@ntu.edu.tw
- ▶ Feel free to interrupt me if you have any questions

Lab Schedule

Time & Location

- ▶ Monday
13:20–14:10
- ▶ Boya 408
- ▶ One week behind
theory class

Weekly Topics

Week	Date	Topic
1	2/23	No class
2	3/2	Introduction to R
3	3/9	Descriptive Statistics
4	3/16	Visualization of Data
5	3/23	Discrete Random Variables
6	3/30	Continuous Random Variables
7	4/6	No class (Holiday)
8	4/13	Sampling Distribution
9	4/20	Estimation
10	4/27	Hypothesis Testing I
11	5/4	Hypothesis Testing II
12	5/11	Hypothesis Testing III
13	5/18	ANOVA
14	5/25	Correlation & Simple Regression
15	6/1	Categorical Data Analysis
16	6/8	No class (Final Exam)

Rules of Lab Session

- ▶ No food or drink allowed!
- ▶ Lab Report: 9% of total grade
 - ▶ 8 reports this semester
 - ▶ Available on NTU COOL
 - ▶ Deadline: Following Sunday at 23:59
 - ▶ Plagiarism → 0% for all involved
- ▶ It should be done during lab session



R and RStudio

R is a programming language used for statistical computing and data visualization.

RStudio is an integrated development environment (IDE) that provides a user-friendly interface for working with R.

- ▶ Free
- ▶ Continuous improvement (R-4.5.3 now)
- ▶ Low barrier to coding
- ▶ Available at:
<https://posit.co/download/rstudio-desktop/>



Installation

Click on the link: <https://posit.co/download/rstudio-desktop/>
Download R **FIRST** before RStudio!

1: Install R

RStudio requires R 3.6.0+. Choose a version of R that matches your computer's operating system.

R is not a Posit product. By clicking on the link below to download and install R, you are leaving the Posit website. Posit disclaims any obligations and all liability with respect to R and the R website.

1

DOWNLOAD AND INSTALL R

2: Install RStudio

macOS 14+ (macOS 14 and higher)

2

DOWNLOAD RSTUDIO DESKTOP FOR MACOS 14+

Size: 666.46 MB | [SHA-256: 8B7FE821](#) | Version:
2026.01.1+403 | Released: 2026-02-19

Download R



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The Comprehensive R Archive Network

Download and Install R

Precompiled binary distributions of the base system and contributed packages, **Windows and Mac** users most likely want one of these versions of R:

- [Download R for Linux \(Debian, Fedora/Redhat, Ubuntu\)](#)
- [Download R for macOS](#)
- [Download R for Windows](#)

R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above.

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R for macOS

This directory contains binaries for the base distribution and of R and packages to run on macOS. R and package binaries for R versions older than 4.0.0 are only available from the [CRAN archive](#) so users of such versions should adjust the CRAN mirror setting (<https://cran-archive.r-project.org>) accordingly.

Note: Although we take precautions when assembling binaries, please use the normal precautions with downloaded executables.

R 4.5.2 "[Not] Part in a Rumble" released on 2025/10/31

Please check the integrity of the downloaded package by checking the signature: `pkgutil --check-signature R-4.5.2-arm64.pkg` in the *Terminal* application. If Apple tools are not available you can check the SHA1 checksum of the downloaded image:
`openssl sha1 R-4.5.2-arm64.pkg`

Latest release:

For Apple silicon (M1,2,...) Macs:

[R-4.5.2-arm64.pkg](#)

SHA1=
hash: 1cc0b3d78bc3b3857c6bf3128a9d414b130d938e
(ca. 97MB, notarized and signed)

For older Intel Macs:

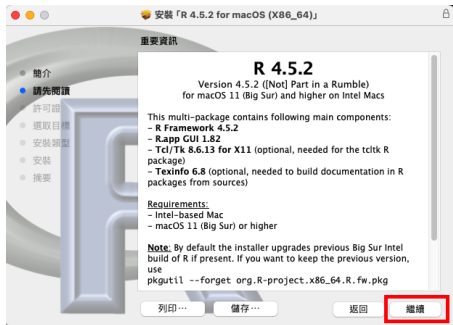
[R-4.5.2-x86_64.pkg](#)

SHA1=
hash: 0184504a11da63b26cc31f91a812e5456d523e0e
(ca. 100MB, notarized and signed)

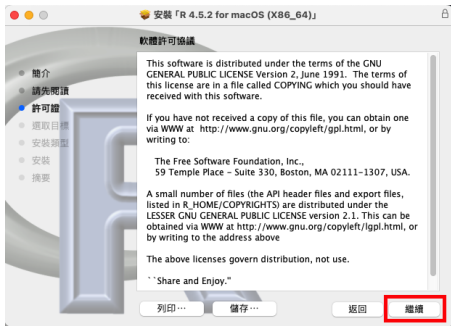
R 4.5.2 binary for macOS 11 (Big Sur) and higher, signed and notarized packages.

Contains R 4.5.2 framework, R.app GUI 1.82, Tcl/Tk 8.6.12 X11 libraries and Texinfo 6.8. The latter two components are optional and can be omitted when choosing "custom install", they are only needed if you want to use the `tcltk` R package or build package documentation from sources.

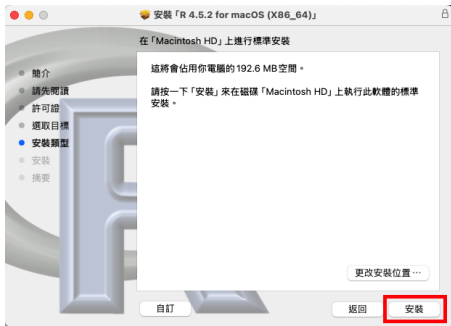
Download R



Download R



Download R



Download RStudio

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DOWNLOAD AND INSTALL R

2: Install RStudio

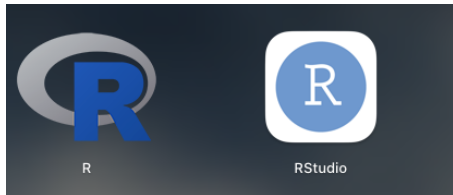
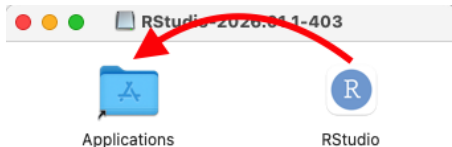
macOS 14+ (macOS 14 and higher) ▾

DOWNLOAD RSTUDIO DESKTOP FOR MACOS 14+

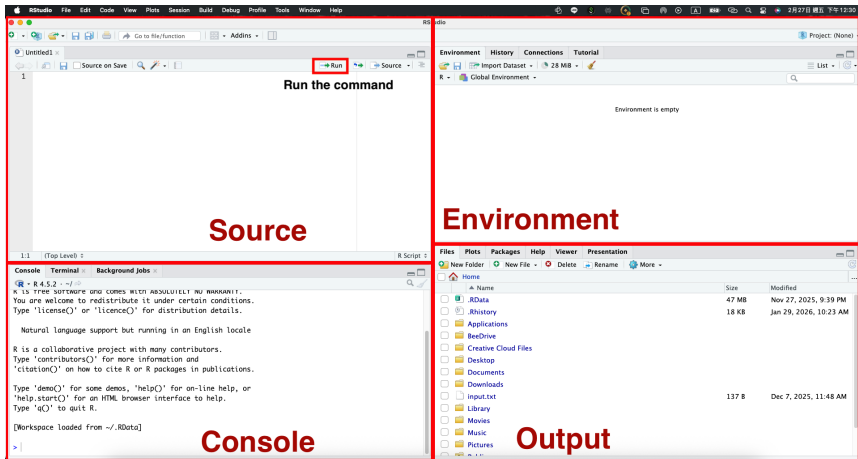
Size: 666.46 MB | SHA-256: 8B7FE821 | Version:
2026.01.1+403 | Released: 2026-02-19

Download RStudio

Check the application to see whether it was successfully installed

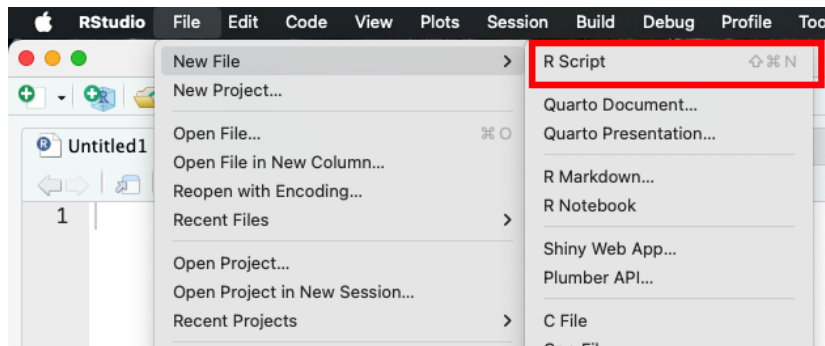


RStudio Interface



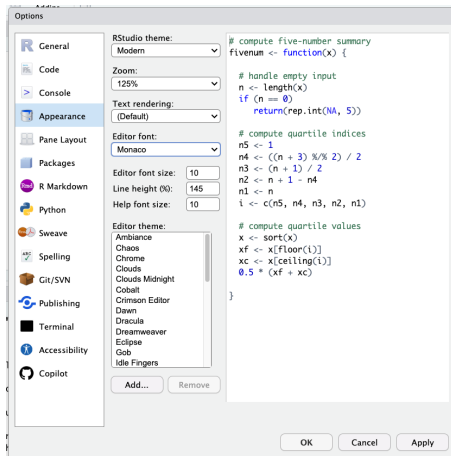
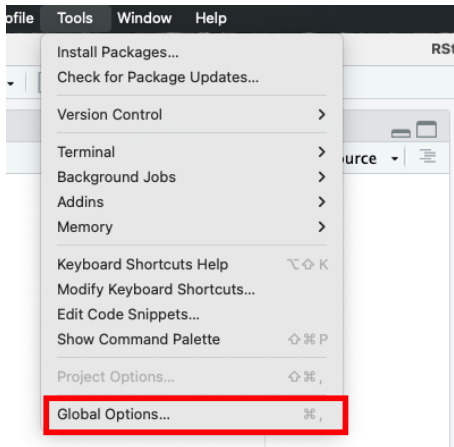
How to create a new R Script

File → New file → R Script



Change the Appearance

Tools → Global Option → Appearance



The Basics: Calculations

- R will do math for you
- How to run a command:
 - ▶ Type Type command into the console and type `Enter`
 - ▶ Type the command in the script and type
 - ▶ `Ctrl` `Enter` (Windows)
 - ▶ `Command` `Enter` (macOS)

Basic Arithmetic Operations

The basic arithmetic operators are:

- ▶ + (addition)
- ▶ - (subtraction)
- ▶ * (multiplication)
- ▶ / (division)
- ▶ ^ (exponentiation)
- ▶ %/% (quotient)
- ▶ %% (modulus)

```
> 1 + 1
[1] 2
> 2026 - 2025
[1] 1
> 3 * 2
[1] 6
> 16 / 20
[1] 0.8
> 2 ^ 10
[1] 1024
> 31 %/% 5
[1] 6
> 31 %% 5
[1] 1
```

Basic Arithmetic Function

Functions accept **parameters** (in parentheses) and return a value.

The basic arithmetic functions are:

- ▶ $\log(x)$: logarithms base e of x
- ▶ $\exp(x)$: Exponential of x
- ▶ $\text{abs}(x)$: absolute value of x
- ▶ $\text{sqrt}(x)$: square root of x

```
> log(2)
[1] 0.6931472
> log10(1000)
[1] 3
> log(exp(3))
[1] 3
> abs(-3)
[1] 3
> sqrt(9)
[1] 3
```

The Basic: Variables

- ▶ Variables are named containers for data.
- ▶ The assignment operator in R is: `<-` or `=`
- ▶ Variable names can start with a letter
 - ▶ Just not a number by itself.
 - ▶ Examples: `result`, `x1`
- ▶ R is case sensitive (i.e. `Result` $\neq$ `result`)

```
> x <- 3
> y = 4
> z = 5
> x + y * z
[1] 23
> rm(x)
> x
```

Error: object 'x' not found

Practice

Use R to calculate the following.

Try assigning values to variables first, then perform the calculation

1.

$$\frac{46 \times 628}{71}$$

2.

$$7 \times (2^8 + 56)$$

3.

$$\frac{\log(|10 - 5| + 1) + \sqrt{3^2 + 5^2} - e^{3-5}}{(3^3 + 5^3 + 1)}$$